

RESEARCH REPORT

Research Overview

Research Identity

Institution	City College of New York
Department	Electrical Engineering
Degree Program	Doctor of Philosophy (PhD)
Advisor	Professor Yi Sun
Report Period	August 2025 – June 2026
Primary Research Domain	KAN in SMLM

Core Research Question

Can Kolmogorov-Arnold Networks (KANs) — architectures grounded in the Kolmogorov-Arnold representation theorem — serve as mathematically principled, interpretable, and parameter-efficient replacements for conventional CNNs and MLPs in the task of emitter detection and localization in Single-Molecule Localization Microscopy?

Main Papers for KAN

The following papers are mainly papers that have been read for exploring KAN.

Papers

Reference	Title / Description	Role in Research
Liu et al., 2024	KAN: Kolmogorov-Arnold Networks	Primary KAN architecture paper; basis for all implementations
Liu et al., 2025	KAN 2.0: KANs meet Science	Extended KAN theory; symbolic regression and science applications
Kolmogorov, 1957	On representation of continuous functions of many variables (Doklady)	Mathematical foundation theorem
Arnold, 1963	On functions of three variables (Doklady)	Arnold refinement of KA theorem
Blealtan, 2024	EfficientKAN: Efficient PyTorch KAN implementation (GitHub)	Vectorized B-spline basis; basis for custom ~48x speedup implementation
Bodner et al., 2024	Convolutional Kolmogorov-Arnold Networks	CKAN extension to image data
Shukla & Karniadakis, 2024	Physics-informed KAN	KANs for PDEs; informs physics-informed framing
Qiu et al., 2024	ReLU-KAN	Efficient KAN variants
Azam & Akhtar, 2024	Suitability of KANs for Computer Vision	KANs in biomedical imaging context

3. Research Timeline

All activities are listed chronologically from August 2025, organized by research milestones, architectural developments, and professional deliverables.

Period	Task	Details
Aug-Sep 2025	-Initial KAN-SMLM Exploration -Installation & Benchmarking -Single-Emitter Pipeline (V1–V5)	First investigation of KAN for SMLM localization. Derived mathematical framework: KAN representation theorem applied to PSF-to-coordinate mapping. Studied B-spline basis functions and convergence properties compared to MLPs. Attempted installation of the package in python. Run the codes of the initial KAN paper and run my code for the dataset on our lab on github- Built first KAN-SMLM pipeline for M=1 (single emitter) localization on 7×7 pixel frames at 100nm/pixel. Implemented patch-based sliding window approach with 28×28 resized inputs.
Oct–Nov 2025	Dataset Infrastructure pyKAN Implementation	Dataset generator for multiple emitters in python. Installed and tested pyKAN. Explored pure B-spline activation configuration.
Nov 2024		
Nov–Dec 2024	Custom PyTorch KAN (V6–V8)	Implemented custom B-spline KAN in pure PyTorch using torch.einsum for vectorized basis evaluation. Verified architectural purity (no nn.Linear layers). Validated equivalence with pyKAN forward pass. Achieved ~48x training speedup.
Dec 2024	Paper Writing (V1)	Wrote first version of KAN-SMLM paper modeled after Nature article structure. Included mathematical formulations (KA theorem, PSF model, loss functions). Submitted to Professor Sun for review.
Jan 2025	PureKANMultiArityLayer	Designed and debugged PureKANMultiArityLayer incorporating pairwise tensor-product B-spline bases for second-order feature interactions. Fixed einsum index errors. Analyzed parameter counts across model variants.
Jan 2025	Multi-Emitter Extension (V9–V10)	Extended pipeline to M=1–5 emitters. Implemented x-coordinate sorting for permutation handling (M=1–3). Adopted Hungarian algorithm matching for M=4–8. Trained detector KAN (M=0–8 classification) and per-M localizer KANs.
Jan–Feb 2025	RMSMD Implementation	Implemented Root Mean Square Minimum Distance (RMSMD) evaluation metric per Prof. Sun's publications. Distinguished from RMSD/Hungarian-RMSE. Computed symmetric nearest-neighbor distances for variable emitter counts.

Feb 2025	Mode Collapse Diagnosis & Fix	Diagnosed mode collapse in pure KAN V10 for $M \geq 3$: activations drifting outside B-spline grid range $(-3, 3)$, causing linear extrapolation and mean-output collapse. Fixed by adding <code>nn.LayerNorm</code> between hidden KAN layers. Created V10_fixed with checkpoints named <code>best_localizer_M{m}_V10_fixed.pth</code> .
Feb–Mar 2025	CNN+KAN Hybrid (V11)	Designed V11: CNN front-end (2 conv blocks, BatchNorm, ReLU) extracts spatial features from $3 \times 7 \times 7$ input (intensity + CoordConv coordinate maps), producing 288-dim feature vector fed to KAN head for sub-pixel coordinate regression. Resolved mode collapse absent in V11.
Mar 2025	Permutation Problem Study	Studied the fundamental permutation/partition problem in multi-emitter localization. Understood why emitter indexing is mathematically necessary: without established correspondence (partition), sample RMSE is undefined. Analyzed Hungarian algorithm and x-sorting as partition estimation methods.
Mar 2025	Abstract & Survey Writing	Began doctoral examination paper: KAN-Loc survey targeting qualifying exam. Wrote and revised abstract collaboratively. Established paper scope: SMLM biological foundations, prior DL approaches, KAN theory, and proposed two-stage KAN-Loc framework.
Apr 2025	Survey Paper Sections 2–6	Wrote Sections 2–6 of doctoral survey: SMLM biological/optical foundations, computational challenges, DL methods in SMLM, KAN architecture theory, and proposed KAN-Loc framework. Managed 26 IEEE-format references with priority placement of Prof. Sun's RMSMD and RMSMD-P papers.
Apr 2025	Ground Truth & Dataset Validation	Generated and validated ground truth files for synthetic 2D Gaussian dataset. Confirmed Dataset-0 parameters (no rotation, no offset) vs. Dataset-1. Corrected coordinate system conventions across pipeline.
Apr–May 2025	Inference & Visualization Pipeline	Built complete inference scripts for V10 and V11 in Google Colab. Wrote per-frame .txt result files following professor-specified naming convention (2DGauss_xy_Frame#_KAN.txt). Created HTML architecture visualizations (dark background, animated neural network diagrams).
May 2025	Model Evaluation & Reporting	Ran full RMSMD evaluation on V10_fixed and V11 across $M=1-5$. Drafted formal technical report email to Professor Sun summarizing results. Created architecture summary tables for both models.
May–Jun 2025	Permutation Invariance Deep Study	Deepened understanding of why RMSMD was designed to avoid partition problem, and where it fails (saturation under large errors). Studied Prof. Sun's RMSMD-P paper on partition-based quality metrics. Connected to Hungarian matching used in training loss.

Jun 2025	Advanced Statistics: PCA Assignment	Completed HW5 for Advanced Statistics course: PCA/EOF analysis of ISCCP 2004 global climate data (TA, WV, WP). Implemented data loading with Fortran column-major reshape, bad data masking, combined TA+WP PCA, EOF spatial maps, PC time series, and day reconstruction at 90% variance threshold.
Jun 2025	Resume & Fellowship Materials	Improved PhD resume summary for AI/ML research positions. Prepared fellowship interview materials. Wrote Google Research internship cover letter.
Jun–Aug 2025	Research Summary to Professor	Drafted and refined multiple academic research summaries for Prof. Sun. Documented NSF SCH fellowship grant work. Connected SMLM research to AI/advanced data science program objectives.
Aug–Sep 2025	Coursework: B-Splines & Robotics	Implemented B-spline polynomial fitting (degree 5) in Python from theory (Cox-de Boor recursion) and using SciPy. Solved robotics coursework: 3D coordinate frame transformations, 4×4 homogeneous transformation matrices, Denavit-Hartenberg parameters across kinematic chains.
Sep 2025	Coursework: ML & Neural Networks	Completed ML coursework: perceptron algorithm analysis (XOR non-separability), backpropagation gradient computation via chain rule, CNN-based fake news detection (RNN+Keras), algorithmic trading ANN improvements (Bollinger Bands, Williams %R, expanded architecture).
Sep–Oct 2025	Data Engineering Project	Built full ETL pipeline: multi-engine web search scraping (Selenium, BeautifulSoup), Tesseract OCR on screenshots, text relevance scoring, MySQL database ingestion, Flask web interface for aggregated results.
Oct 2025	Pure KAN Implementation Validation	Addressed concerns about 'Pure KAN' classification. Confirmed custom PyTorch implementation is correctly described as Pure KAN: B-spline basis only, no nn.Linear, no MLP branch. Documented 48x speedup rationale vs. pyKAN library.
Nov–Dec 2025	Paper Draft (Nature-level)	Wrote complete first-version paper for Nature-level submission based on KAN-Loc algorithm. Included mathematical formulas, PSF physics, B-spline implementation details, RMSMD evaluation framework, and benchmark comparisons.
Jan–Feb 2026	Dataset Generation for Geometric Patterns	Generated synthetic 2D Gaussian dataset with emitters in geometric patterns across varying separation distances (d=25nm to d=150nm). Validated ground truth files. Computed RMSMD metrics across separation distances.
Mar 2026	Proposal: Sparse Deep Unrolling	Developed separate research proposal: Sparse Deep Unrolling for Overlapping Fluorophore Localization. Studied ISTA/ADMM unrolling, LISTA architecture (Gregor & LeCun), physics-informed Poisson loss. Aligned with NSF SCH program objectives.

Apr 2026	Doctoral Exam Survey Paper (Final)	Finalized KAN-Loc doctoral survey paper with complete 26-reference bibliography (IEEE format). Wrote all sections including paper structure, SMLM review, DL review, KAN theory, and KAN-Loc framework. Updated citations: KAN → ICLR 2025 Oral, KAN 2.0 → Phys. Rev. X.
Apr 2026	Citation Verification & Reference Management	Verified publication status of all 26 references. Updated KAN paper status to ICLR 2025 Oral. Confirmed KAN 2.0 published in Physical Review X vol.15. Confirmed remaining arXiv-only references with current status.
May–Jun 2026	Permutation Invariance (Professor's Question)	Answered professor's question about the fundamental reason for emitter indexing in KAN: established that the partition problem makes sample RMSE mathematically undefined without correspondence assignment. Studied Prof. Sun's RMSMD-P paper in depth.

Programming Languages, Tools & Environment

Language	Libraries / Frameworks	Used For
Python (primary)	PyTorch, NumPy, SciPy, Pandas, Matplotlib, PIL,tiffle	
Python	pyKAN	Initial KAN library before transitioning to custom implementation
Python	TensorFlow/Keras	
Python	scikit-learn	
Python	scipy.optimize (linear_sum_assignment)	
MATLAB	Dataset Generation	
JavaScript	HTML/CSS, Canvas API	neural network architecture visualizations
Google Colab	Drive mounting, GPU runtime	Primary training and inference environment for all models

Development Environment

- Primary platform: Google Colab (GPU runtime, T4/G4/A100)
- Secondary platform: VS Code/ Cursor